

LEADERSHIP BRIEFING · SEPTEMBER 2026

Mooring the wave-flume test articles

How we designed a mooring that keeps the buoys in view without changing the motion we are there to measure

SUMMARY

The bottom line

Ready to buy

The specification (rev C.1) is complete. It waits only on the test lab confirming seven facility assumptions.

Measures true

The mooring shifts the buoys' rocking period by at most 1.4 %, inside the tolerance we set first.

Stays in view

In the largest waves, stiffer cords keep the articles inside the 1 m camera window.

One rig, two sets of elastic cords: soft for the measurement tests, stiff for the big-wave tests.

THE TEST

Three test articles, one wave flume

Article	What it is
Single buoy	One 1:50-scale buoy with a heave plate
Cluster	4 buoys hinged to a common hub, set at 45°
Platform	16 buoys in a 4 × 4 array, 2.1 m across

The flume

3.66 m wide, 2.7 m deep

Regular waves up to 0.5 m high, 1.4 to 3.5 s apart

Goal: measure the motion to validate FloatSim, our simulation tool

THE PROBLEM

The mooring has two jobs that pull apart

Hold position

- Stay inside the camera window: 0.3 m up-flume to 1.0 m down-flume
- Stay clear of the flume walls
- Never let a cord go slack in the measurement waves

Stay out of the way

- Shift the natural rocking period by less than 1.0 to 1.4 %
- Keep the calm-water lean at 1° or less
- Leave the up-and-down motion unchanged

Stiff cords hold position; soft cords stay out of the way. The design had to do both.

PROCESS

How we designed it

- 1 Agree the pass/fail criteria first.** Seven criteria, set before any design work: position, lean, rocking period, slack, loads.
- 2 Simulate every option in FloatSim.** Attachment height, cord stiffness and tension, for all three articles.
- 3 Choose by a written rule.** The rule picks the design, not case-by-case judgement.
- 4 Verify the choice.** Full FloatSim wave runs of the chosen design, including the worst waves.
- 5 Record every decision.** Specification, design basis and figures are all generated from the same simulation files.

House rule: every number is recomputed from the record, never copied, and any disagreement is written down.

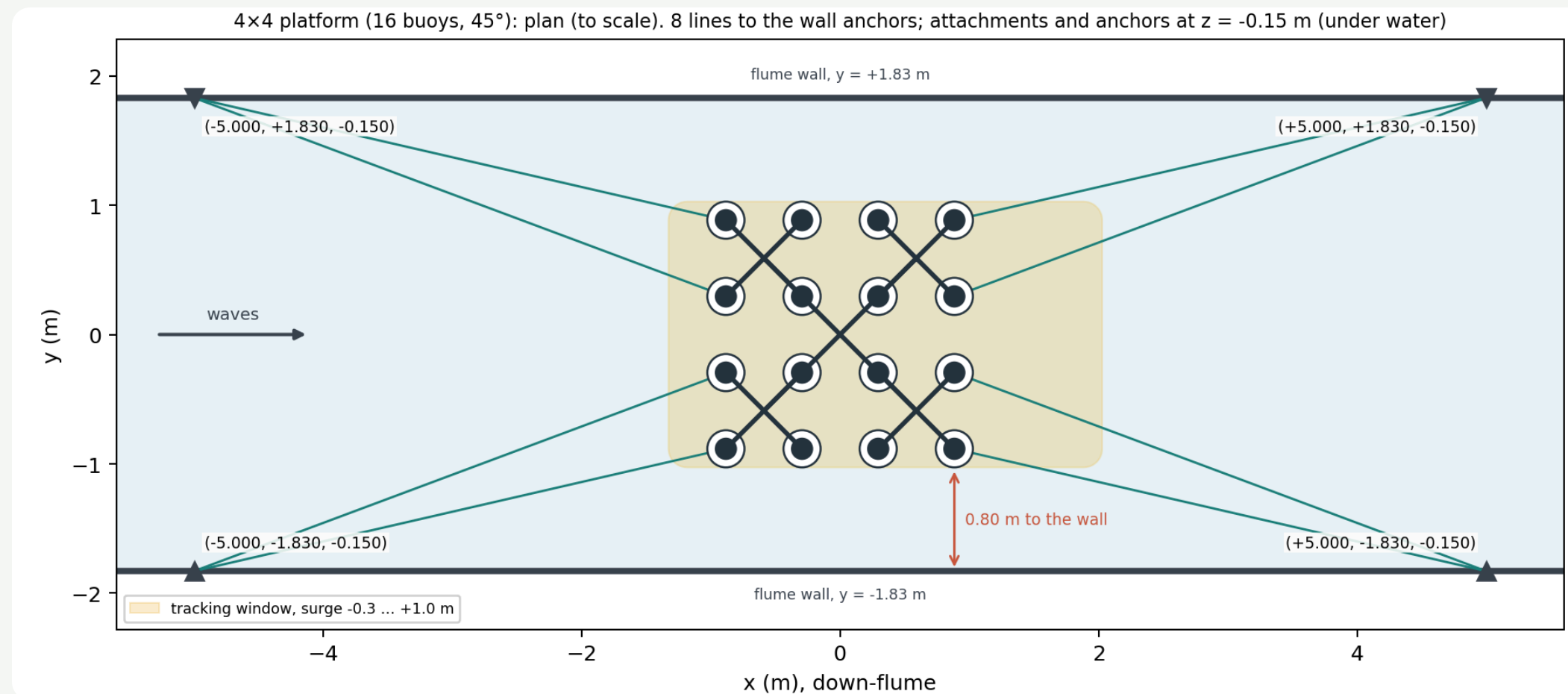
HOW THE DESIGN EVOLVED

Four revisions, each fixing the last one

Revision	Design	What we learned
Rev A	Cords above water, at the hinge line	Shifted the rocking period ~9 %, 6–9× the tolerance: withdrawn
Rev B	Cords 15 cm under water, low tension	Measures true, but 0.5 m waves push articles 3.4–3.5 m, out of view
Rev C	Two cord sets on the same anchors	Soft set measures true, stiff set holds position: both jobs done
Rev C.1	Daily check for rubber creep	Cords relax ~13 % in a day: check and re-tension before testing

THE DESIGN

Elastic cords to underwater wall anchors



Plan view of the 16-buoy platform, to scale (from the specification)

5 m

to the anchors, up- and down-flume,
on the side walls

15 cm

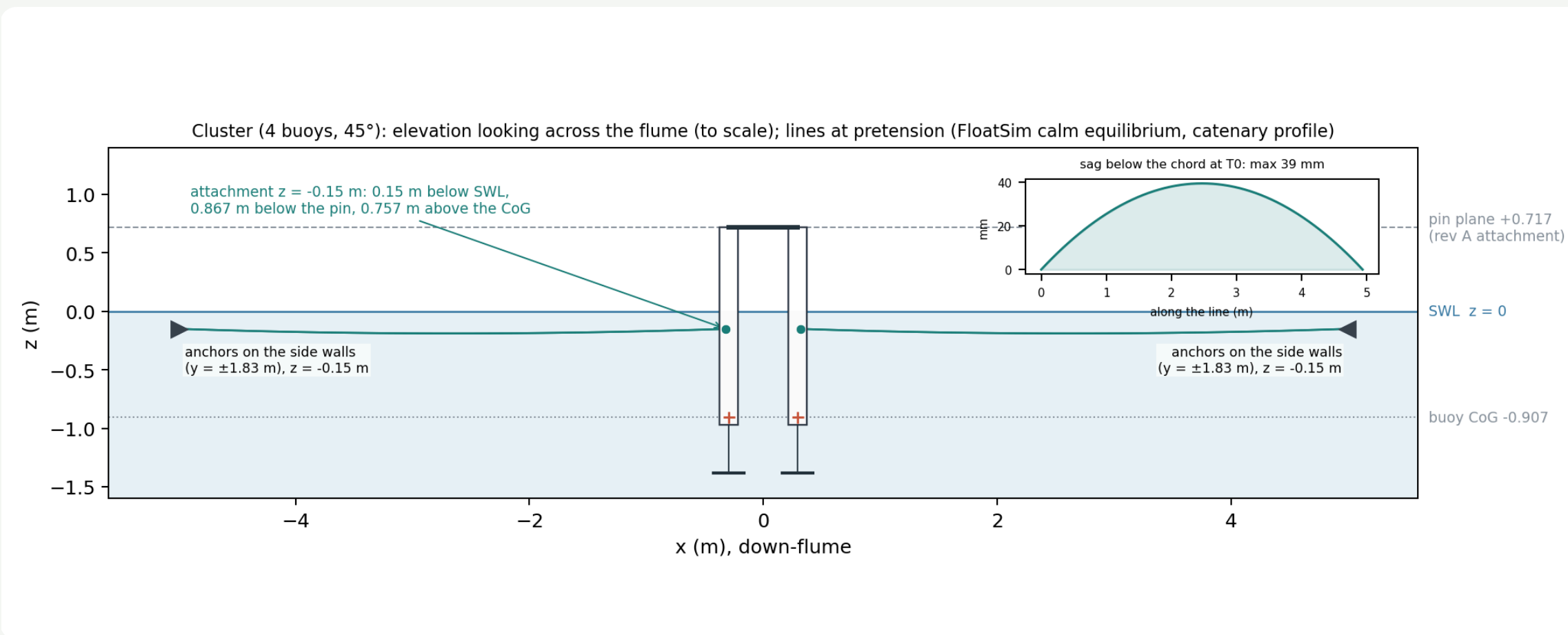
below the water line, where the cords
attach (single buoy: 50 cm)

0.8 m

clearance from the platform to each
wall

THE KEY INSIGHT

Why the cords attach below the water line



Side view of the 4-buoy cluster, to scale, cords at rest

High up, a cord levers on the rocking: a ~9 % period shift.

Just below the water, the lever almost vanishes: under 1 %.

Trade-off: a steady 1° lean, now our tension gauge.

The mooring barely changes the motion

1.4 %

the most the rocking period moves (single buoy; cluster and platform 0.7 %), each inside its tolerance

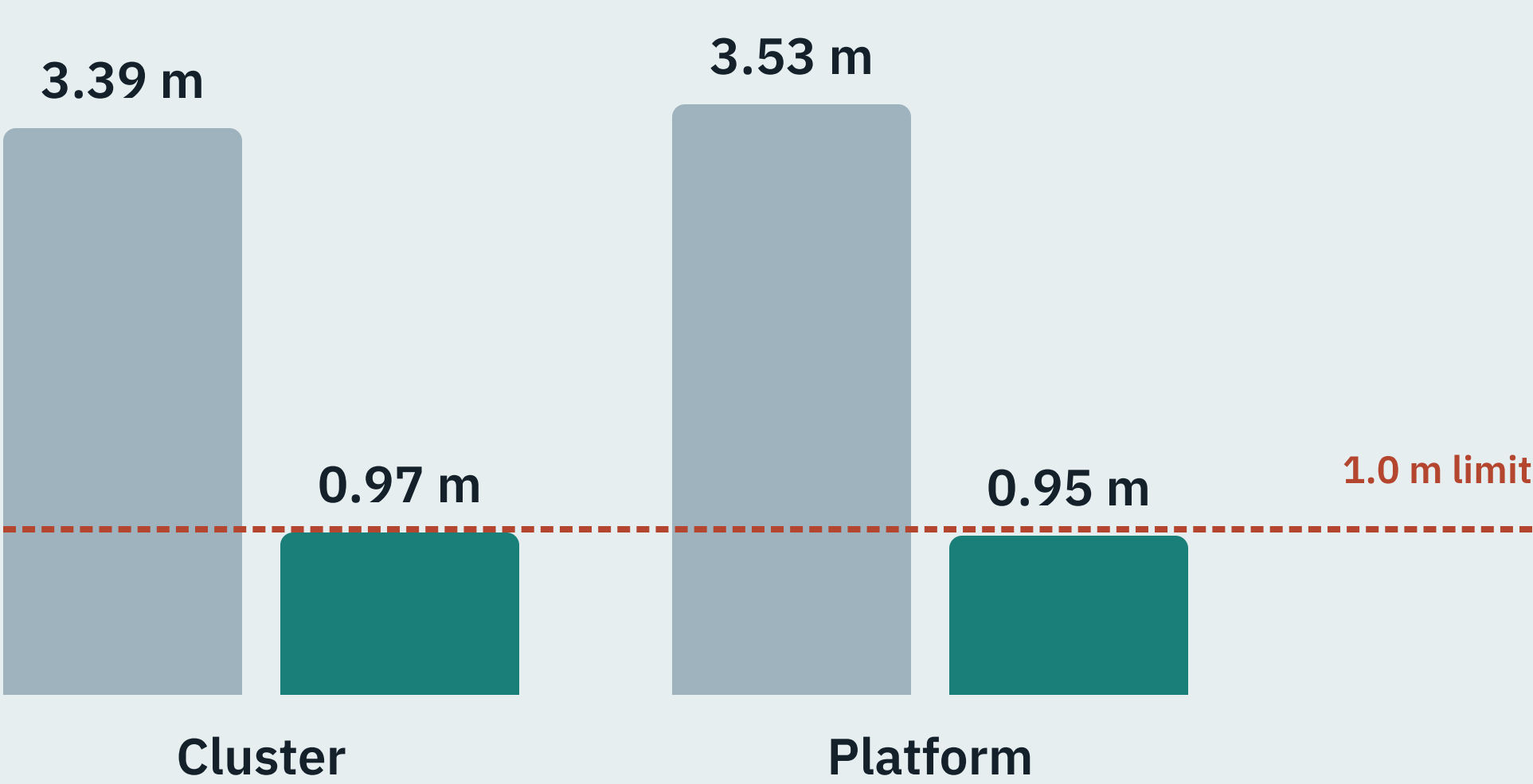
≤ 5 %

heave difference, moored vs free, where the buoys move most (up to 9 % in one mid-length wave)

1°

steady lean from the cord tension: confirmed acceptable, and used to check it

Stiff cords keep big-wave tests in view



- Soft cords (measurement set)
- Stiff cords (big-wave set)

In the **largest waves (0.5 m)**, soft cords let the articles drift 3.4–3.5 m, out of view. Stiff cords hold them at 0.95–0.97 m, inside the limit.

In the **measurement waves (up to 0.12 m)**, the soft cords hold drift under 7 cm.

HOW THE TEST RUNS

One rig, two cord sets

Soft set

Measurement tests, waves up to 0.12 m

- The article moves almost as if free
- Cords stay taut in every measurement wave

Stiff set

Load and survival tests, 0.2 to 0.5 m

- Cords 5–6× stiffer, same anchors
- Holds the article inside the 1 m window

Swapping sets: same anchors, same tension at rest, same 1° lean. So a quick pull test, required before every big-wave series, confirms which set is on.

DECISIONS

Key decisions and why

Decision	Why
Keep a small cord tension	Taut cords behave the way the simulator models them; slack, flapping cords do not
Keep the single buoy's tension	Its collar uses the tension to keep the buoy from spinning
Two cord sets, not one compromise	No single cord can both measure true and hold position in 0.5 m waves
Size stiff set to the strict limit	Average drift plus wave motion, not just the average, stays inside 1.0 m

HARDWARE

What we will buy

Anchors

Four per article, on the walls under water, each rated at least 300 N (largest need: 267 N)

Cords

Elastic shock cord that works under water: soft and stiff sets, one set for the single buoy

Buoy collar

A slender spreader, rod 25 mm or thinner. A solid disk would add ~21 kg of water mass

Daily check

Rubber relaxes ~13 % a day: check the tension, or the 1° lean, before the first run

OPEN RISKS

What could still change the design

Risk	How we handle it
Lab facility assumptions not yet confirmed	Confirm with the test lab before buying anything
Simulator is approximate at large tilts	Treat those cases as indicative; upgrade planned
Single buoy: 3 big-wave cases unpredicted	Run last, raise waves gradually, stop on any spin
Cord snap loads are not simulated	3× margin on loads; load-cell pull test at install

NEXT STEPS

From specification to the flume

- 1 Confirm the facility assumptions with the test lab.** Seven items, including underwater wall anchors rated at 300 N.
- 2 Procure cords and anchors.** Both cord sets; confirm the cord's creep rate with the supplier.
- 3 Finish the verification.** A refined model of the buoys and the full 442-case simulation matrix.
- 4 Upgrade the simulator for large tilts.** Needed to predict the resonant and largest-wave cases directly.

[Decision requested: confirm, e.g. approval to procure once the test lab confirms]

Supporting material

Document	What it holds
Mooring specification rev C.1	One-page summary, line tables, figures, installation checks, procedure
Design basis	Every decision, the evidence behind it and the disagreements settled
Pretension evaluation	Why the cords keep their tension (options compared)
3D motion viewer	Moored vs free motion of each article in 8 wave cases